

I

$$-10 + I_1 - 2I_1 + V_{TH} = 0$$

$$V_{TH} = 10 + I_1 \quad \text{--- (1)}$$

$$-10 + I_1 + I_1 + 15 = 0$$

$$2I_1 = -5$$

$$I_1 = -2.5$$

$$V_{TH} = 7.5$$

$$R_{TH} = \frac{V}{I}$$

$$b I_1 = 2 \times \frac{1}{2} = -I$$

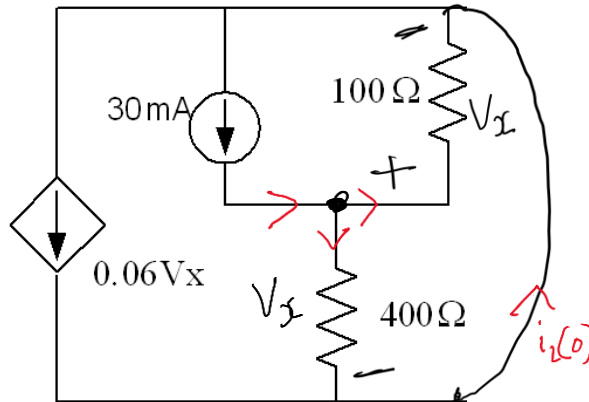
$$\text{Inner loop } V = I + I + \frac{I}{2} \quad V = \frac{I}{2}$$

$$R_{TH} = \frac{V}{I} = 0.5 \Omega$$

$$P_{\max} = \frac{V_T^2}{4R_L} = \frac{7.5^2}{4 \times 0.5} = 28.125W$$

2

Before $t=0$

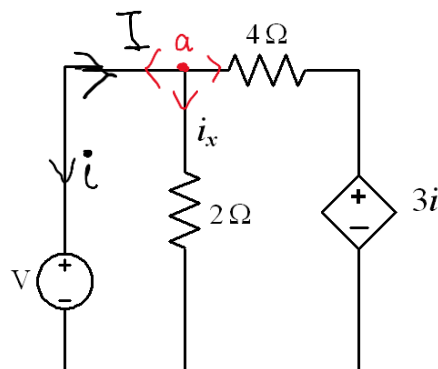


$$\frac{V_x}{100} + \frac{V_x}{400} = 30 \times 10^{-3}$$

$$\frac{5V_x}{400} = 30 \times 10^{-3}$$

$$V_x = 2.4V$$

$$i_L(0) = \frac{V_x}{400} + 0.06V_x = 0.006 + 0.144 = 0.15A$$



$$R_{TH} = \frac{V}{I} \quad i = -I$$

$$\text{KCL at } a \quad -I + \frac{V}{2} + V + \frac{3I}{4} = 0$$

$$\frac{3V}{4} = \frac{1}{4}I \quad R_{TH} = \frac{V}{I} = \frac{1}{3} \Omega$$

$$L/R = 0.5/1/3 = 1.5 \text{ sec}$$

$$i(t) = I(0)e^{-t/\tau} = 150 \times 10^{-3} e^{-t/1.5} A$$

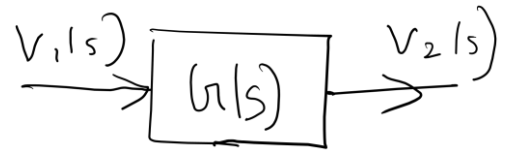
$$v_L(t) = L \frac{di}{dt} = 0.5 \left(\frac{-1}{\tau} \right) I(0) e^{-t/\tau}$$

$$i_x = \frac{v_L}{2} =$$

$$i_x(t) = -\frac{150 \times 10^{-3}}{6} e^{-t/1.5}$$

3.

$$G(s) = \frac{V_2(s)}{V_1(s)}$$



$$V_1(t) = 10u(t) - 10u(t - 5 \times 10^{-6})$$

$$V_1(s) = \frac{10}{s} - \frac{10}{s} e^{-5 \times 10^{-6} s}$$

$$G(s) = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}$$

$$Z_2(s) = R_3$$

$$Z_1(s) = \frac{(R + Ls) \times (R + \frac{1}{Cs})}{R + Ls + R + \frac{1}{Cs}}$$

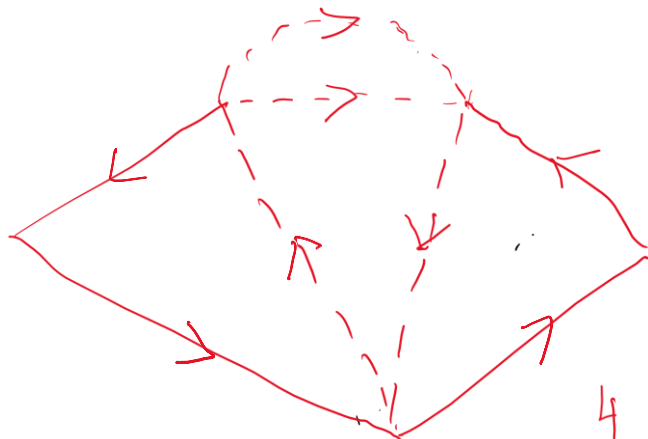
$$V_2(s) = \bar{L}^{-1} (G(s) V_1(s))$$

4.

$$\boxed{Z I_c = E_L} \text{ where } Z = B Z_b B^T \quad E_L = -B_E^T E_b$$

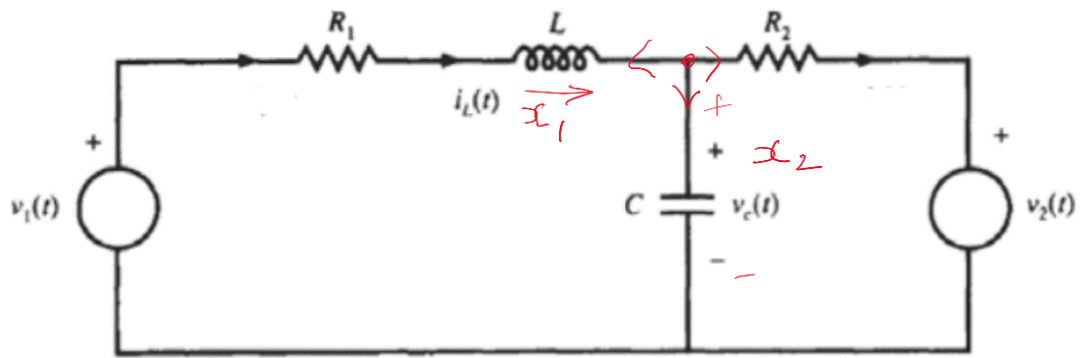
I_c = vector of link currents.

Find Z



4 fund. circuits

5.



$$\begin{aligned} \text{KVL} \quad -V_1 + R_1 x_1 + L \dot{x}_1 + x_2 &= 0 & \dot{x}_1 &= -\frac{R_1}{L} x_1 - \frac{1}{L} x_2 + \frac{1}{L} V_1 \quad (1) \\ \text{KCL} \quad \frac{x_2 - V_2}{R_2} + C \dot{x}_2 - x_1 &= 0 & \dot{x}_2 &= \frac{1}{C} x_1 - \frac{1}{CR_2} x_2 + \frac{1}{CR_2} V_2 \quad (2) \end{aligned}$$

$$\begin{aligned} y_1 &= R_1 x_1 \\ y_2 &= L \dot{x}_1 = -R_1 x_1 - x_2 + V_1 \\ y_3 &= x_2 - V_2 \end{aligned} \quad \rightarrow \quad \begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \\ \dot{y}_3 \end{bmatrix} = \begin{bmatrix} R_1 & 0 \\ -R_1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

6

Solve the state equation $\dot{x}_1 = x_2$
 $\dot{x}_2 = -2x_1 - 3x_2 + 1$ to find x_1 at time $t = 2$ sec

$$X(s) = (sI - A)^{-1} x(0) + (sI - A)^{-1} B U(s)$$

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad u = 1 \quad U(s) = \frac{1}{s}$$

$$(sI - A)^{-1} x(0) = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 \\ -2 \end{bmatrix}$$

$$(sI - A)^{-1} B U(s) = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} 1 \\ s \end{bmatrix} \frac{1}{s}$$

```
ts=[0 2];
x0=[1;0];
[t,x]=ode45('midterm',ts,x0);
a=[t,x];
[m,n]=size(a);
a(m,2)
```

```
function c = midterm(t,x)
c=[x(2);-2*x(1)-3*x(2)+1];
```

Finding inverse transform, $x_1(t) = 0.5 + e^{-t} - 0.5e^{-2t}$. At $t=2$, $x_1 = 0.6261$

a = 0	1.0000	0
0.0001	1.0000	-0.0001
0.0001	1.0000	-0.0001
0.0002	1.0000	-0.0002
.	.	.
.	.	.
1.9569	0.6313	-0.1213
1.9677	0.6300	-0.1202
1.9785	0.6287	-0.1192
1.9892	0.6274	-0.1181
2.0000	0.6261	-0.1170
t	x_1	x_2